

Day 10 – Service Version Enumeration Report

Trios Cyber Internship

Assignment: Service Version Enumeration **Date:** 10 September 2026 **Target:** Metasploitable 2 **Target IP:** 192.168.56.101 **Environment:** Authorized Local VirtualBox Laboratory **Tool:** Nmap 7.99 **Status:** Completed

1. Objective

The objective of this assessment was to perform service and version enumeration against an authorized local laboratory target using Nmap.

The assessment focused on:

- Identifying open TCP services.
- Detecting the service and software versions running on those ports.
- Running Nmap's safe default NSE scripts.
- Reviewing service-specific information returned by the scripts.
- Building a technology and service inventory.
- Comparing the discovered services with the expected Metasploitable 2 laboratory attack surface.
- Identifying notable or anomalous services and configurations.
- Documenting the findings professionally.

No exploitation was performed as part of this assignment.

2. Scope and Authorization

Testing was performed exclusively against the intentionally vulnerable Metasploitable 2 virtual machine in the local VirtualBox laboratory environment.

Item	Details
Target	Metasploitable 2
IP Address	192.168.56.101
Scanner	Kali Linux
Scanner IP	192.168.56.102
Network	VirtualBox Host-Only Network
Authorization	Local cybersecurity training laboratory
Testing Type	Service/version enumeration and safe default scripting

The assessment was limited to the authorized laboratory target.

3. Methodology

The following Nmap command was used:

```
nmap -sV -sC 192.168.56.101
```

Nmap Options

Option	Description
-sV	Performs service and version detection
-sC	Runs Nmap's default NSE scripts
Target IP	Specifies the authorized laboratory system

The scan was executed from Kali Linux and the complete output was preserved as raw evidence in:
logs/service-version-scan.txt

4. Scan Result

The target was reachable and identified as an active Metasploitable 2 system.

The scan reported:

- **1 host up**
- **977 closed TCP ports**
- **23 open TCP ports**
- Multiple service and software versions
- Additional information from Nmap default NSE scripts

The target's MAC address was identified as an Oracle VirtualBox virtual NIC.

5. Discovered Service Inventory

Port	Service	Version / Details
21/tcp	FTP	vsftpd 2.3.4
22/tcp	SSH	OpenSSH 4.7p1 Debian 8ubuntu1
23/tcp	Telnet	Linux telnetd
25/tcp	SMTP	Postfix smtpd
53/tcp	DNS	ISC BIND 9.4.2
80/tcp	HTTP	Apache httpd 2.2.8 (Ubuntu) DAV/2
111/tcp	RPCbind	RPC #100000
139/tcp	NetBIOS/SMB	Samba 3.X-4.X
445/tcp	SMB	Samba 3.0.20-Debian
512/tcp	rexec	netkit-rsh rexecd
513/tcp	rlogin	login service
514/tcp	rsh	Netkit rshd
1099/tcp	Java RMI	GNU Classpath grmiregistry
1524/tcp	Bind shell	Metasploitable root shell
2049/tcp	NFS	Versions 2-4
2121/tcp	FTP	ProFTPD 1.3.1
3306/tcp	MySQL	5.0.51a-3ubuntu5
5432/tcp	PostgreSQL	8.3.0-8.3.7
5900/tcp	VNC	Protocol 3.3
6000/tcp	X11	Access denied
6667/tcp	IRC	UnrealIRCd
8009/tcp	AJP13	Apache Jserv Protocol v1.3
8180/tcp	HTTP	Apache Tomcat 5.5

6. Screenshot Evidence – Service and Version Detection

The following screenshot provides evidence of the Nmap service and version detection results.

Figure 1: Nmap -sV -sC service and version enumeration results.

7. Default NSE Script Findings

Nmap's default scripts provided additional service-level information.

```
Session Actions Edit View Help
(yuvarej@kali) ~/Tris-Cyber-Internship/Day-10-Service-Version-Enumeration
$ cat logs/service-version-scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 20:17 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.102
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu004-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp_commands: metasplitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_SSL2 DES_192_EDE3_CBC_WITH_MD5
|_SSL2 DES_64_CBC_WITH_MD5
|_SSL2 RC4_128_WITH_MD5
|_SSL2 RC4_128_EXPORT40_WITH_MD5
|_SSL2 RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2 RC2_128_CBC_WITH_MD5
|_ssl-date: 2026-09-10T14:48:16+00:00; +1s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
```

Figure 1: Nmap Service Version Scan

FTP

The scan reported that anonymous FTP login was allowed on port 21.

SMTP

The SMTP service exposed capabilities including:

- PIPELINING
- VRFY
- ETRN
- STARTTLS
- Other SMTP extensions

The default scripts also identified legacy SSLv2 support.

RPC and NFS

RPC enumeration identified:

- rpcbind
- NFS
- mountd
- nlockmgr
- status

This demonstrated that additional RPC-associated services were available on the target.

SMB

Samba was detected on ports 139 and 445.

The default scripts identified:

- Computer name: metasplitable
- Domain: localdomain
- Samba version: 3.0.20-Debian
- Authentication level: user

- Challenge-response support
- SMB message signing disabled
- SMB2 negotiation failure

MySQL

The MySQL service on port 3306 was identified as version 5.0.51a-3ubuntu5.

VNC

The VNC service on port 5900 reported protocol version 3.3 and VNC authentication.

Other Script Information

The scan also returned information related to SSH host keys, HTTP server details, Tomcat, X11, and other services.

8. Screenshot Evidence – Default Script Findings

```

Session Actions Edit View Help
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-10 20:17 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0013s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-syst:
|_STAT:
|_FTP server status:
|_Connected to 192.168.56.102
|_Logged in as ftp
|_TYPE: ASCII
|_No session bandwidth limit
|_Session timeout in seconds is 300
|_Control connection is plain text
|_Data connections will be plain text
|_vsftpd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_2048 50:5b:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu0804-base.localdomain/organizationName=OC0SA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2018-03-17T14:07:45
|_Not valid after: 2018-04-10T14:07:45
|_smtp-command: metasplitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITIME, DSN
|_ssl-v2:
|_SSLv2 supported
|_ciphers:
|_SSL2 DES_192_EDE3_CBC_WITH_MD5
|_SSL2 DES_64_CBC_WITH_MD5
|_SSL2 RC4_128_WITH_MD5
|_SSL2 RC4_128_EXPORT40_WITH_MD5
|_SSL2 RC2_128_CBC_EXPORT40_WITH_MD5
|_SSL2 RC2_128_CBC_WITH_MD5
|_ssl-date: 2026-09-10T14:48:16+00:00; +1s from scanner time.
53/tcp    open  domain       ISC BIND 9.4.2
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|_program version  port/proto service

```

Figure 2: Nmap Default Script Findings

Figure 2: Nmap default NSE script findings, including SMB security information and other service-level observations.

9. Expected Services vs. Notable Findings

Metasploitable 2 is intentionally designed as a vulnerable cybersecurity training system. Therefore, the presence of numerous legacy services and outdated software is expected within this laboratory.

Expected Services

The following services were consistent with the known laboratory attack surface:

- FTP
- SSH

- Telnet
- SMTP
- DNS
- HTTP
- SMB
- RPC/NFS
- MySQL
- PostgreSQL
- VNC
- IRC
- Tomcat
- AJP

Notable Security Observations

Port	Observation	Security Relevance
21	Anonymous FTP enabled	Unauthenticated file access may expose resources
23	Telnet exposed	Legacy remote access protocol that does not provide modern encrypted administration
25	SSLv2 support detected	Legacy cryptographic protocol
445	SMB signing disabled	Reduces protection against certain SMB-related attacks
512–514	rexec/rlogin/rsh exposed	Legacy remote administration services
1524	Bind shell identified	Highly sensitive service intentionally present in the lab
5900	VNC protocol 3.3	Legacy remote desktop protocol
6000	X11 exposed	Remote display service identified, although access was denied

These observations represent security concerns that would require review on a production system.

10. Screenshot Evidence – Service Inventory and Anomalies

Figure 3: Service inventory and anomaly assessment based on the Nmap results.

11. Anomaly Assessment

No unexpected services were identified relative to the known purpose of the Metasploitable 2 training environment.

However, the scan identified several services and configurations that would be considered significant security concerns in a modern production environment.

The most notable observations were:

1. Anonymous FTP access was enabled.
2. Telnet was exposed.
3. Legacy remote shell services were exposed on ports 512–514.
4. A bind shell was exposed on port 1524.
5. Multiple outdated software versions were identified.
6. SMB message signing was disabled.
7. Legacy SSLv2 support was detected.
8. Multiple database and remote administration services were exposed.

```

(yuvraj@kali) ~/Triforce-Cyber-Internship/Day-10-Service-Version-Enumeration
$ cat ANOMALY-SUMMARY.md
# Day 10 - Service Version Enumeration: Expected Services and Anomaly Review

## Target

* **Target:** Metasploitable 2
* **IP Address:** 192.168.56.101
* **Environment:** Authorized Local VirtualBox Laboratory
* **Scan Command:** nmap -sV -sC 192.168.56.101

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## 1. Expected Lab Services

The target is an intentionally vulnerable Metasploitable 2 virtual machine. Therefore, the presence of multiple legacy and intentionally insecure services is expected within this authorized training environment.

Examples of expected services identified during the scan include:

| Port | Service | Detected Version / Details |
|-----|-----|-----|
| 21 | FTP | vsftpd 2.3.4 |
| 22 | SSH | OpenSSH 4.7p1 |
| 23 | Telnet | Linux telnetd |
| 25 | SMTP | Postfix smtpd |
| 53 | DNS | ISC BIND 9.4.2 |
| 80 | HTTP | Apache httpd 2.2.8 |
| 139/445 | SMB | Samba 3.0.20 |
| 2049 | NFS | NFS versions 2-4 |
| 3306 | MySQL | MySQL 5.0.51a |
| 5432 | PostgreSQL | PostgreSQL 8.3.x |
| 5900 | VNC | Protocol 3.3 |
| 6667 | IRC | UnrealIRCd |
| 8188 | HTTP | Apache Tomcat/5.5 |

These services are consistent with the intentionally vulnerable design of the Metasploitable 2 laboratory target.

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## 2. Notable Findings

Several services exposed by the target were notable because they use legacy protocols, outdated versions, weak configurations, or services that would normally require justification in a production environment.

| Port | Finding | Observation |
|-----|-----|-----|

```

Figure 3: Service Inventory and Anomalies

Because this system is specifically designed for security training, these conditions are expected and were documented for educational analysis.

12. Security Recommendations

If equivalent findings were identified on a production system, recommended actions would include:

- Disable unnecessary services.
- Replace Telnet and legacy remote-shell services with secure alternatives.
- Disable anonymous FTP unless explicitly required and controlled.
- Upgrade outdated server software and operating systems.
- Disable obsolete cryptographic protocols such as SSLv2.
- Review SMB security configuration and enable appropriate message signing where required.
- Restrict database services to trusted networks and hosts.
- Restrict remote administration services through appropriate network controls.
- Review exposed services regularly using authorized vulnerability-management processes.
- Apply least-privilege and network-segmentation principles.

These recommendations are provided as defensive security guidance and were not implemented against the laboratory target during this assessment.

13. Evidence Files

Evidence	Location
Raw Nmap scan	logs/service-version-scan.txt
Service/version screenshot	screenshots/01-nmap-service-version-scan.png
Default-script screenshot	screenshots/02-nmap-default-script-findings.png
Service/anomaly screenshot	screenshots/03-service-inventory-anomalies.png
Anomaly assessment	ANOMALY-SUMMARY.md
Practical worksheet	SERVICE-VERSION-ENUMERATION-WORKSHEET.md

14. Commands Executed

Command	Purpose
<code>nmap -sV -sC 192.168.56.101</code>	Service/version detection and safe default NSE scripts
<code>nmap -sV -sC 192.168.56.101 \ tee logs/service-version-scan.txt</code>	Execute the scan while preserving raw output
<code>grep -E '^[0-9]+\tcp[[:space:]]+open' logs/service-version-scan.txt</code>	Extract the discovered open TCP service inventory
<code>cat logs/service-version-scan.txt</code>	Review saved scan evidence
<code>cat ANOMALY-SUMMARY.md</code>	Review the service/anomaly assessment

15. Learning Outcomes

This assessment demonstrated practical understanding of:

- Nmap service and version detection.
- Nmap default NSE scripts.
- Service fingerprinting.
- Port-to-service mapping.
- Version identification.
- Interpretation of service-specific enumeration results.
- Building a technology/service inventory.
- Comparing observed services against an expected laboratory attack surface.
- Identifying security-relevant configurations.
- Maintaining raw evidence and supporting documentation.

16. Conclusion

The Day 10 Service Version Enumeration assessment was successfully completed against the authorized Metasploitable 2 laboratory target.

Nmap identified **23 open TCP services** and provided detailed service/version information. The default NSE scripts supplied additional information about FTP, SMTP, RPC/NFS, SMB, MySQL, VNC, and other services.

The discovered services were consistent with the intentionally vulnerable design of the Metasploitable 2 training environment. Several services and configurations were nevertheless identified as security concerns that would require remediation or restriction in a production environment.

The assessment was performed strictly within the authorized local laboratory environment. No exploitation or intrusive testing was performed.

Assessment Status: Completed